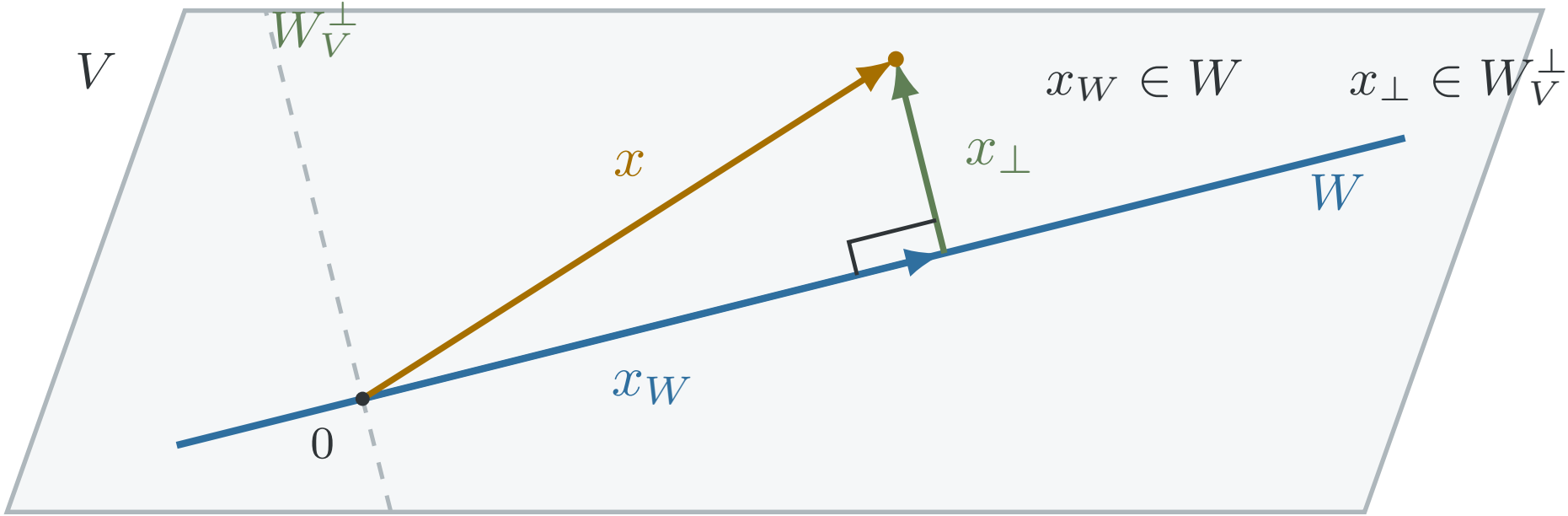


$$x = x_W + x_\perp$$

orthogonal projection + residual

$$V = W \oplus W_V^\perp$$

the decomposition is unique

$$\dim V = \dim W + \dim W_V^\perp$$